

Local Mean-Force and Harmonic/Manifold Refinement Specification

Stage 11E3

mdstats

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Scope

Stage 11E3 refines the position-derived statistical candidates from Stage 11E2 with force evidence from the Stage 11E0b sample catalog. It does not create, delete, merge, or label spatial sites. A candidate remains present when force data are unavailable, inadmissible for PMF use, locally under-sampled, singular, or inconsistent with a stable harmonic model.

The implementation owner is:

`mdstats.analysis.density.force_refinement`

The stage consumes one source-compatible triple:

`FrameworkAlignedIonSampleCatalog`
`PeriodicSpeciesDensityEstimate`
`DensityAttractorCatalog`

and returns one immutable `ForceRefinementCatalog`.

Borrowed methods and package-specific constructions

The conditional-average-force interpretation follows Darve and Pohorille (2001), DOI 10.1063/1.1410978. The local linear force-matching interpretation follows Noid et al. (2008), DOI 10.1063/1.2938860. Complete-system blocking for correlated uncertainty follows Flyvbjerg and Petersen (1989), DOI 10.1063/1.457480.

The following are mdstats-specific constructions:

- exact source binding across E0b, E1, and E2 signatures;
- retaining every E2 attractor regardless of force-evidence status;
- matched use of the E1 periodized Gaussian and image-truncation certificate;
- explicit Cartesian-to-fractional covector conversion before score comparison;
- support intersection with the E1 support mask;
- lifted periodic residence charts bound to one E2 attractor;
- separate density anchor, force center, curvature class, and chart-containment status;

- separate ordinary fit, covariance diagnostic, and density-force residual; and
- fail-closed serialization and resource preflight.

Coordinate and force contract

Let the registered periodic domain use row-vector fractional coordinates $\mathbf{q}$ and cell matrix H :

$$\mathbf{x} = \mathbf{q}H.$$

A registered Cartesian force covector $\mathbf{F}_x$ is transformed to the fractional coordinate measure by work invariance:

$$\mathbf{F}_q = \mathbf{F}_x H^T, \quad \mathbf{F}_x d\mathbf{x}^T = \mathbf{F}_q d\mathbf{q}^T.$$

The E1 density score $d_q \log p$ and $\mathbf{F}_q$ are therefore compared in the same covector measure. The analysis metric is not applied until an orthonormal local chart is required.

If $G = LL^T$ is the E1 analysis metric and $\delta_y = \delta_q L$, then local force covectors transform as

$$\mathbf{F}_y = \mathbf{F}_q L^{-T}.$$

Matched-kernel conditional mean force

Only the E0b pmf_force evidence subset is admissible. For query node $\mathbf{q}_g$, represented-time weights w_n , forces $\mathbf{F}_{q,n}$, and the same periodized Gaussian $K_h^{\mathbb{T}^3}$ used by E1,

$$\bar{\mathbf{F}}_q(\mathbf{q}_g) = \frac{\sum_n w_n K_h^{\mathbb{T}^3}(\mathbf{q}_g - \mathbf{q}_n) \mathbf{F}_{q,n}}{\sum_n w_n K_h^{\mathbb{T}^3}(\mathbf{q}_g - \mathbf{q}_n)}.$$

The field stores:

- conditional force covector;
- conditional force covariance;
- local effective sample size;
- support mask;
- optional complete-system block standard error;
- represented ion time and force-sample count; and
- E0b/E1 source signatures.

A node is force-supported only when it also belongs to the E1 support mask and passes the configured local effective-sample threshold. Unsupported field values are finite zeros; the support mask is authoritative.

Density-score comparison

At declared constant temperature T and for PMF-admissible equilibrium sampling,

$$\bar{\mathbf{F}}_q(\mathbf{q}) \approx k_B T d_q \log p(\mathbf{q}).$$

For each E2 attractor, the result records the norm of the residual at the representative logical node when both channels are supported. This residual is a diagnostic. It does not overwrite either field or project the force field onto a conservative subspace.

Lifted residence chart

Samples assigned to attractor i use the E2 basin owner at the nearest periodic logical node. Relative positions are lifted by

$$\delta_q = (\mathbf{q} - \mathbf{q}_i) - \text{rint}(\mathbf{q} - \mathbf{q}_i).$$

A lift is accepted only when every component lies strictly inside the periodic half-cell cut. The E3 fit chart is the union of the E2 local chart and accepted residence lifts. This extension is recorded by the fit sample indices; it does not alter the E2 topology.

Symmetric local force fit

For point candidates and resolved extended candidates, fit

$$\mathbf{f}_n = \mathbf{b}_i - \delta_{y,n} K_i + \epsilon_n, \quad K_i = K_i^T,$$

with represented-time weighted least squares. The nine fitted parameters are three intercept components and six independent stiffness components. The result retains:

- design rank and condition number;
- intercept and symmetric stiffness;
- stiffness eigenvalues and eigenvectors;
- residual covariance;
- parameter standard errors;
- force-defined point center where identifiable; and
- center-within-chart status.

For nonsingular point fits,

$$\delta_y^{(F)} = \mathbf{b}_i K_i^{-1}.$$

The center is not imposed for an extended attractor.

Curvature classes

The force fit reports one of:

```
stable_point  
saddle_or_unstable  
soft_manifold  
flat_or_unresolved  
not_evaluated
```

A stable point requires all stiffness eigenvalues above the declared minimum. A saddle or unstable fit has at least one eigenvalue below the negative threshold. A resolved one-dimensional E2 manifold is `soft_manifold` when two normal directions are restoring and the soft eigenvalue is small relative to the largest stiffness. Otherwise the result remains unresolved.

Residence covariance diagnostic

For all position-admissible samples assigned to the provisional residence basin, compute the represented-time weighted covariance in the same orthonormal chart:

$$\Sigma_i^{(\text{res})} = \text{Cov}(\delta_y \mid B_i).$$

When K_i is positive definite and a constant temperature is declared, also report

$$\Sigma_i^{(\text{harm})} = k_B T K_i^{-1}$$

and the relative Frobenius discrepancy. This is diagnostic only. Basin truncation, anharmonicity, many-ion conditioning, and finite sampling may make the two covariances disagree.

Evidence status

Every E2 attractor receives exactly one `LocalForceRefinement`. Its independent force-evidence status is one of:

```
resolved  
force_unavailable  
pmf_provenance_rejected  
insufficient_local_support  
chart_unresolved  
rank_deficient  
ill_conditioned  
center_outside_chart
```

No non-resolved status deletes the position-derived candidate.

Resource and serialization contract

LocalMeanForceResourcePolicy preflights grid nodes, force samples, Gaussian image terms, workspace, output bytes, and attractor count before allocation. All options, fields, refinements, and catalogs have deterministic SHA-256 signatures and strict replay constructors. Field replay requires numerical values; summary-only dictionaries are not replayable scientific objects.

Non-goals

Stage 11E3 does not:

- reconstruct a global PMF;
- project a raw force field to its conservative component;
- perform Helmholtz-Hodge decomposition;
- infer transition events or residences;
- assign structural or crystallographic site labels;
- estimate barriers or rates; or
- replace the E2 nonparametric basin catalog with harmonic ellipsoids.

Acceptance tests

The focused gate must verify:

1. a synthetic harmonic well recovers center and stiffness;
2. unstable curvature is not labeled a stable point;
3. an extended attractor preserves a soft direction without a point center;
4. missing or PMF-inadmissible force data preserve all spatial attractors;
5. insufficient samples and singular designs lower evidence status;
6. matched fields retain E1 support and block uncertainty;
7. source mismatch and resource excess fail closed;
8. serialization replay is exact and tamper-evident; and
9. public exports and stage metadata are stable.

References

- Darve, E., and Pohorille, A. (2001). *Calculating Free Energies Using Average Force*. Journal of Chemical Physics 115, 9169-9183. DOI: 10.1063/1.1410978.
- Noid, W. G., et al. (2008). *The Multiscale Coarse-Graining Method. I. A Rigorous Bridge between Atomistic and Coarse-Grained Models*. Journal of Chemical Physics 128, 244114. DOI: 10.1063/1.2938860.
- Flyvbjerg, H., and Petersen, H. G. (1989). *Error Estimates on Averages of Correlated Data*. Journal of Chemical Physics 91, 461-466. DOI: 10.1063/1.457480.